March 22

1. ( )
P, Q, R

(a) “ ”
(b) “ Bug 5 ”
(c) “ ”

2. ( )
$A = (P \to Q) \land \neg Q \to \neg P$

(a)
(b)

3. ( )
$PV = \{p_1, p_2, p_3, \ldots\}$ $\{\neg, \lor, \land, \to, \leftrightarrow\}$ { (,) } $PROP$
$PROP$ $\aleph_0$

4. ( )
$L(A)$ A “(” $R(A)$ A “)”
Structural Induction $A \in PROP$ $L(A) = R(A)$

5. ( )
3 Tautologies

(a) $P \to (Q \to P)$
(b) $(P \to Q) \leftrightarrow (\neg Q \to \neg P)$
(c) $(P \lor Q) \to (\neg P \to Q)$

6. ( )
$\hat{v}$ 3

(a) $(P \to (Q \to R)) \to ((P \to Q) \to (P \to R))$
(b) $(\neg P \to \neg Q) \to (Q \to P)$
(c) $((P \to Q) \land (Q \to R)) \to (P \to R)$

7. ( )
$F = (P \lor Q) \land (\neg P \lor R) \land (\neg Q \lor \neg R)$

(a) Satisfiable
(b) True (T) $\hat{v}$

8. ( )
Lec 01 NMOS PMOS XOR $F = (P \lor Q) \land \neg(P \land Q)$

(a) · F “ (NAND)” “ (NOR)”
(b) NAND NOR